

Assignment 02

September 10, 2026

Solutions to exercises 3.2, 3.3, 3.4 for assignment 02, Programmer som Data.

3.2

"Write a regular expression that recognizes all sequences consisting of a and b where two a 's are always separated by at least one b "

A regular expression that matches the above:

$$b^*(a \mid b)(b \mid ba)^*$$

As we interpret the exercise, the empty string ε is not accepted. This is reflected in the middle term $(a \mid b)$, which requires at least one symbol.

The regular expression can be represented by the following NFA:

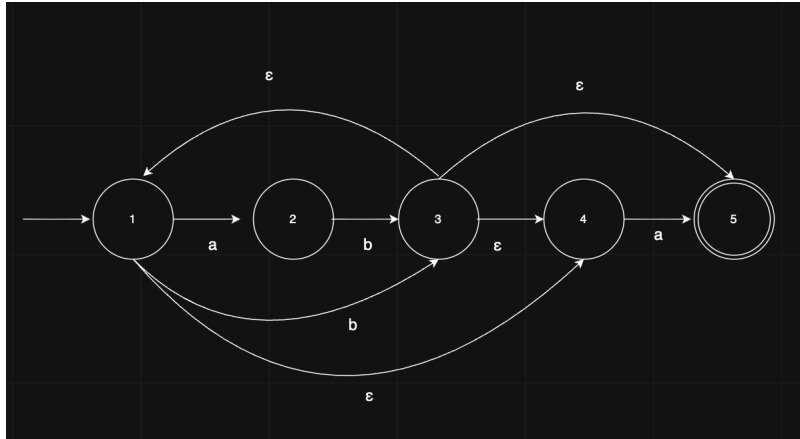


Figure 1: NFA for the regular expression $b^*(a \mid b)(b \mid ba)^*$, with accepting state 5

This NFA can be transformed to the following DFA, as pictured below.

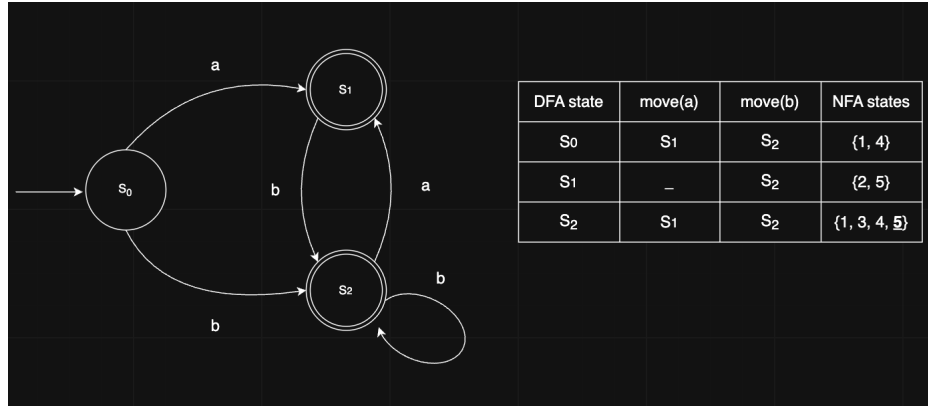


Figure 2: Corresponding DFA and its subset construction table, showing DFA states, transitions, and their corresponding NFA state sets

3.3

The following table shows each step of the right-most derivation of the string:

let z = (17) in z + 2 * 3 end EOF

as well as the production rules applied at each step.

Derivation step	Rule
Main	(start symbol)
⇒ Expr EOF	A
⇒ LET NAME EQ Expr IN Expr END EOF	F
⇒ LET NAME EQ Expr IN Expr PLUS Expr END EOF	H
⇒ LET NAME EQ Expr IN Expr PLUS Expr TIMES Expr END EOF	G
⇒ LET NAME EQ Expr IN Expr PLUS Expr TIMES CSTINT(3) END EOF	C
⇒ LET NAME EQ Expr IN Expr PLUS CSTINT(2) TIMES CSTINT(3) END EOF	C
⇒ LET NAME EQ Expr IN NAME(z) PLUS CSTINT(2) TIMES CSTINT(3) END EOF	B
⇒ LET NAME EQ LPAR Expr RPAR IN NAME(z) PLUS CSTINT(2) TIMES CSTINT(3) END EOF	E
⇒ LET NAME EQ LPAR CSTINT(17) RPAR IN NAME(z) PLUS CSTINT(2) TIMES CSTINT(3) END EOF	C
⇒ LET NAME(z) EQ LPAR CSTINT(17) RPAR IN NAME(z) PLUS CSTINT(2) TIMES CSTINT(3) END EOF	B

3.4

The tree corresponding to the derivation from exercise 3.3:

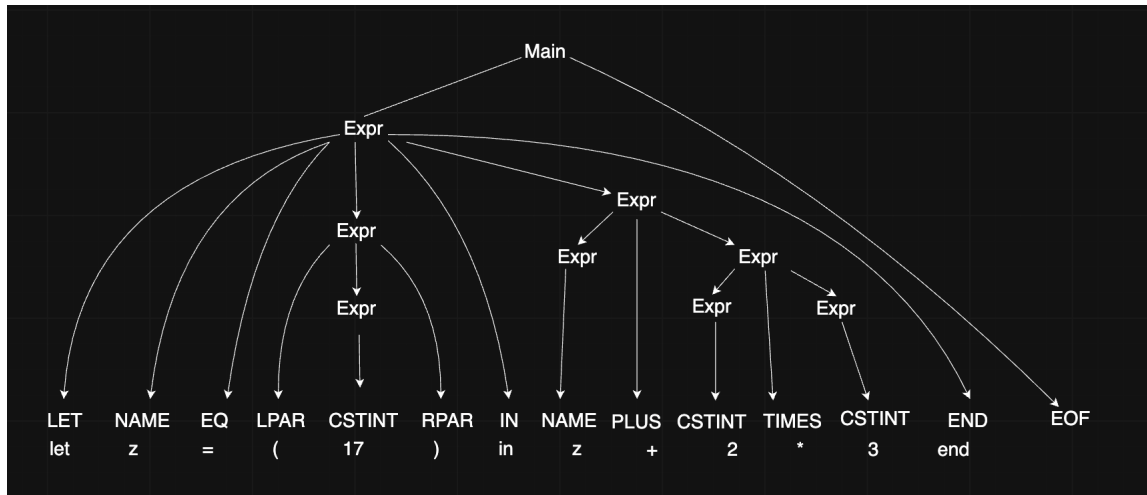


Figure 3: Tree corresponding to the derivation in 3.3